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PENETRATION TESTING REPORT

Field	Detail
Pentester Name (Cybersecurity Professional)	Frank Ivine Marara
Date	18 September 2026
Modules Completed	W2-PM1(Multiple Kali Tools) W2-PM5 (Zenmap Scanning)
Client/Target	1.Networkwalks (secured written permission already) 2.My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: in Progress
Environment	Kali Linux in Oracle VirtualBox

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Introduction

The assessment reviewed DNS information, host discovery, network reconnaissance, and basic web server security indicators. The supplied screenshots show testing from Kali Linux using DNSRecon, nslookup, Nmap through Zenmap, and Nikto. The work focused on identifying exposed infrastructure, reachable hosts, DNS records, web server characteristics, and basic HTTP security-header issues.

Tools Used

Tool	Purpose	Key Findings
WHOIS	Domain registration details	Registrar: GoDaddy; Nameservers: HostGator; Expiry: 2027
DNSRecon	DNS enumeration	MX, TXT, and SRV records found; SPF and Google verification tokens
Nslookup	DNS resolution	Domain resolves to IP 192.232.216.135
Nikto	Web server vulnerability scan	Missing CSP, HSTS, and X-Content-Type-Options headers
WAFW00F	WAF detection	ModSecurity (SpiderLabs) identified

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Whatweb	Web technology fingerprinting	WordPress 7.1.1, Bootstrap 7.1.1, Apache server
Zenmap/Nmap	Network and port scanning	Active hosts: 10.0.0.1–10.0.0.3; Open ports: 135, 445, 7070
Ping Scan	Host discovery	3 hosts up in subnet 10.0.0.0/24

Activities Performed

a) DNS Enumeration

An initial DNSRecon command produced an argument error, as shown in Evidence 1. A corrected DNSRecon run then completed successfully for networkwalks.com. The successful run identified two nameservers, an A record, an MX record, an SPF-related TXT record, and multiple SRV records for autodiscover.

- A record: networkwalks.com resolved to 192.232.216.135.
- Nameservers observed: ns6136.hostgator.com and ns6135.hostgator.com.
- MX record observed: mail.networkwalks.com.
- TXT record observed: an SPF policy containing ip4:50.87.144.87 and an include for websitewelcome.com.
- SRV records observed for _autodiscover._tcp.networkwalks.com pointing to cpanelmaildiscovery.cpanel.net on port 443.
- DNSRecon reported no answer for its DNSSEC query. This observation alone does not prove the complete DNSSEC configuration status.

b) DNS Lookup Verification

nslookup used the local resolver at 192.168.1.1 and returned networkwalks.com as 192.232.216.135. This independently confirms the address shown by the DNSRecon evidence.

c) Internal Host Discovery

Zenmap performed a Ping Scan against 10.0.0.0/24 using the Nmap command `nmap -sn 10.0.0.0/24`. The scan reported three hosts as up: 10.0.0.1, 10.0.0.2, and 10.0.0.3. The scan covered 256 IP addresses and completed in about 3.03 seconds.

d) Internal Network Reconnaissance

Zenmap also ran an Intense Scan against 10.0.0.0/24 using `nmap -T4 -A -v 10.0.0.0/24`. The evidence shows the scan processing the /24 range and reporting many addresses as down. The supplied screenshot does not show the complete service and port results for 10.0.0.1 through 10.0.0.3, so no port-level conclusion is made from this screenshot.

e) Web Server Reconnaissance

Nikto 2.6.0 targeted networkwalks.com at 192.232.216.135 on port 80. The server was identified as Apache, and the output included WordPress-related headers. Nikto also reported

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several missing recommended HTTP security headers and observed responses involving numerous archive-style paths.

- Missing Content-Security-Policy header reported.
- Missing Strict-Transport-Security header reported.
- Missing X-Content-Type-Options header reported.
- A WordPress-related response header was observed.
- Nikto observed cookies being returned from several tested paths, including a `__wpm_client` cookie.
- The evidence does not establish successful access to backup files or sensitive data. The archive-style path responses should therefore be treated as an exposure indicator requiring manual verification.

Risk Analysis

The risk ratings below reflect the evidence supplied in the screenshots. They describe security exposure indicators rather than confirmed compromise.

Finding	Evidence	Risk	Potential Impact	Priority
Missing HTTP security headers	Nikto output	Medium	Reduced browser-side protection against selected web attacks and content-type abuse.	High
HTTP service identified on port 80	Nikto target and server response	Medium	Unencrypted HTTP traffic can expose sessions or content if HTTPS enforcement is incomplete.	High
Potentially exposed archive/backup-style paths	Nikto path responses	High if sensitive files exist	Accidental disclosure of configuration, source, backups, or other sensitive material.	High
DNS information disclosure	DNSRecon and nslookup	Low	Public DNS records reveal hosting, mail, and service infrastructure information.	Medium
Internal hosts discovered	Nmap Ping Scan	Low to Medium	Reachable internal assets	Medium

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			provide a starting point for authorized internal security testing.	
DNSSEC query returned no answer	DNSRecon	Informational	The screenshot does not provide enough evidence to confirm whether DNSSEC is absent or configured incorrectly.	Low

Recommendations

1. Enforce HTTPS across the public website and redirect HTTP traffic to HTTPS.
2. Enable HSTS after confirming HTTPS works correctly across the domain and required subdomains.
3. Add Content-Security-Policy after testing the policy against the WordPress installation and required third-party resources.
4. Add X-Content-Type-Options: nosniff.
5. Review all paths reported by Nikto, especially archive, backup, compressed, certificate, database, and configuration-style filenames. Remove unnecessary files from the web root and block direct access to sensitive files.
6. Review WordPress plugins, themes, and the WordPress core version. Keep all components patched and remove unused components.
7. Review cookie settings and use Secure, HttpOnly, and appropriate SameSite attributes for session or sensitive cookies.
8. Review DNS records and remove obsolete records. Confirm SPF, DKIM, and DMARC policies for mail security.
9. Confirm the intended internal network segmentation. Restrict management services and unnecessary ports between VLANs or network segments.
10. Repeat the assessment after remediation and retain before-and-after evidence.

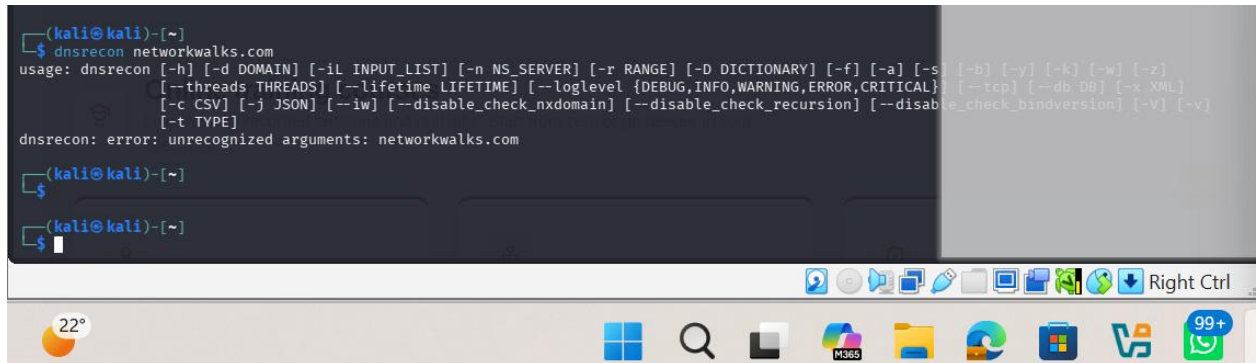
Conclusion

The test confirmed that networkwalks.com is operational and protected by ModSecurity, but several **configuration weaknesses** exist. The supplied evidence shows a successful reconnaissance exercise covering DNS, public web infrastructure, and an internal /24 network. The public target resolved to 192.232.216.135, while the internal discovery scan identified three active hosts in the 10.0.0.0/24 range. Nikto identified Apache and WordPress-related indicators and reported missing HTTP security headers. The strongest follow-up item is a manual review of archive and backup-style paths because the screenshots show responses from those paths, while the evidence does not prove sensitive-file disclosure. The recommended remediation focuses on HTTPS enforcement, HTTP security headers, web-root

- hygiene, WordPress maintenance, cookie security, DNS hygiene, and internal network segmentation.

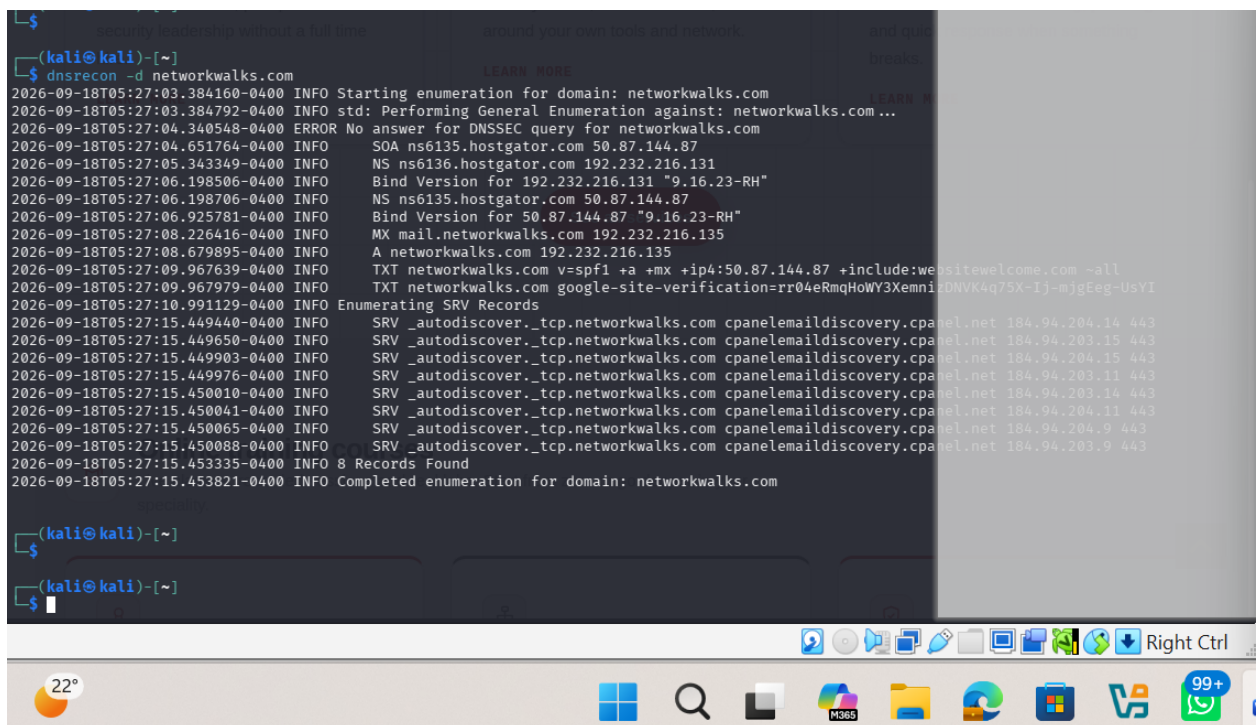
Evidences and Screenshots

Initial DNSRecon command error

A screenshot of a Kali Linux terminal window. The user has entered the command 'dnsrecon networkwalks.com'. The terminal output shows the usage of the dnsrecon tool and an error message: 'dnsrecon: error: unrecognized arguments: networkwalks.com'. The terminal window has a dark background with a light-colored text area. The taskbar at the bottom shows various icons including a weather icon (22°), a search icon, and several application icons like a file manager, a web browser, and a terminal. The system clock shows 'Right Ctrl'.

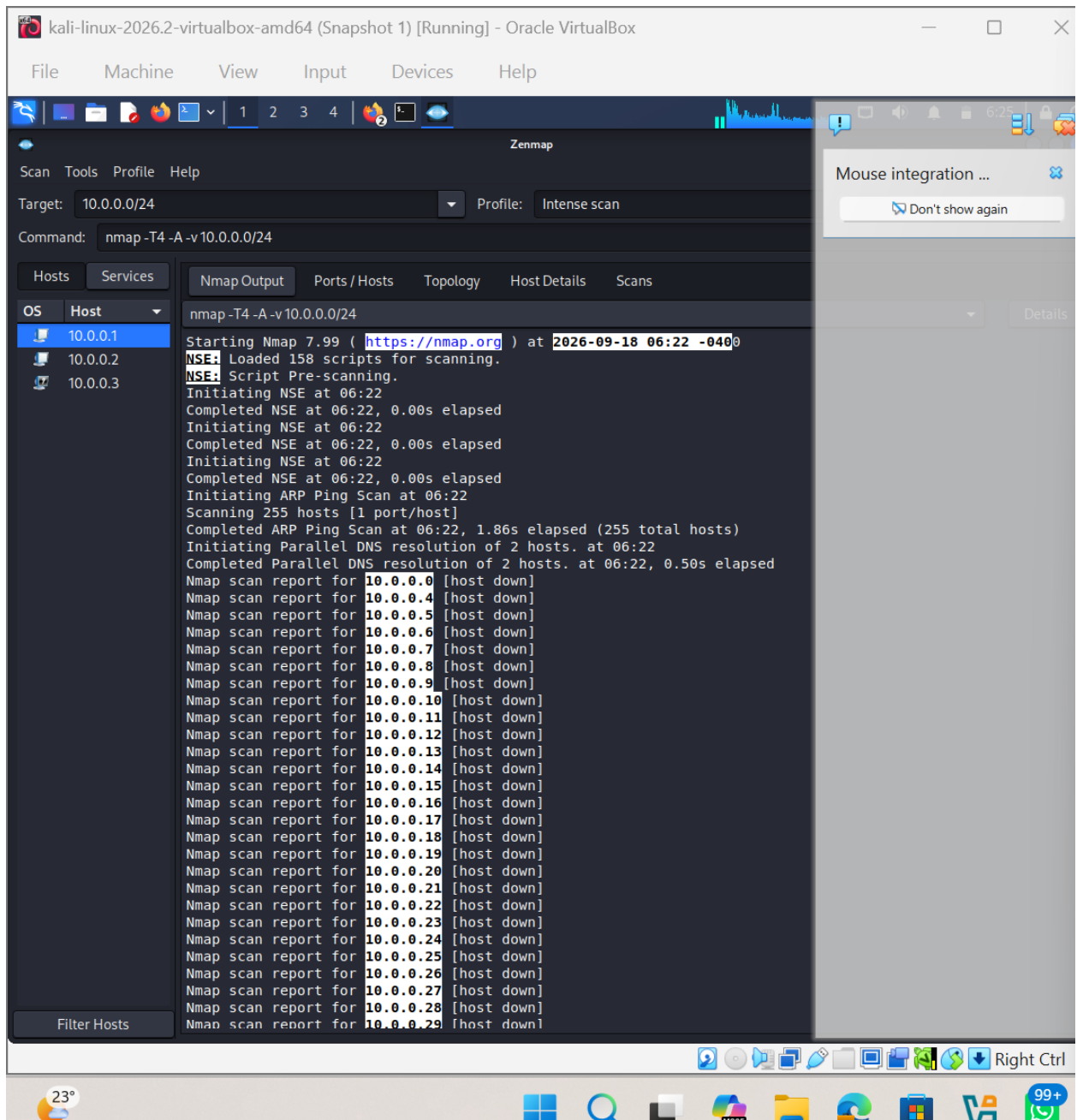
The first DNSRecon attempt returned an 'unrecognized arguments' error, indicating an incorrect command format.

Successful DNSRecon enumeration

A screenshot of a Kali Linux terminal window showing the output of the 'dnsrecon -d networkwalks.com' command. The output is a detailed log of the enumeration process, including information about the domain, the DNS server, and the records found. The records include NS, A, MX, TXT, and SRV records. The terminal window has a dark background with a light-colored text area. The taskbar at the bottom shows various icons including a weather icon (22°), a search icon, and several application icons like a file manager, a web browser, and a terminal. The system clock shows 'Right Ctrl'.

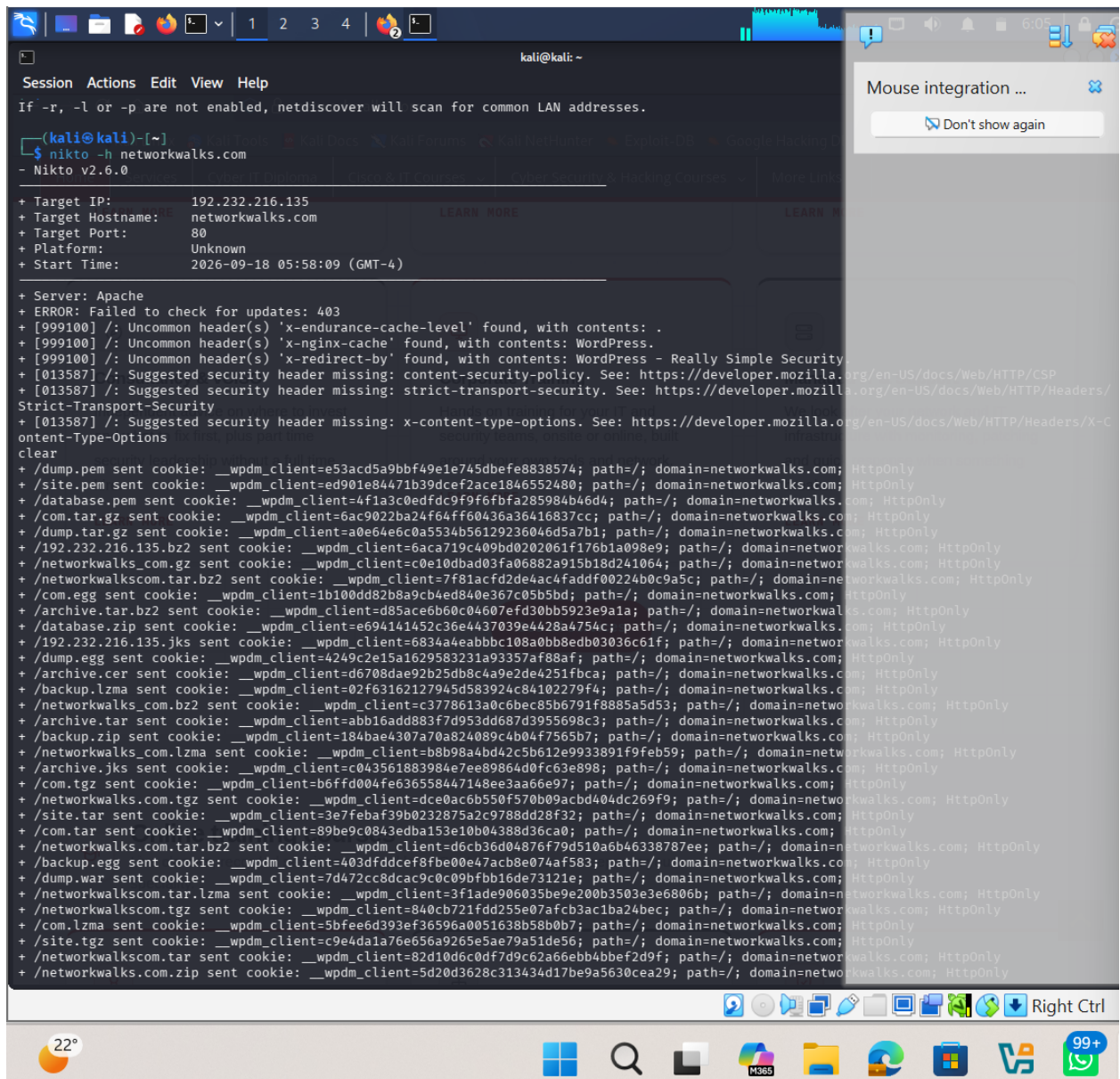
Successful enumeration of DNS records for networkwalks.com, including NS, A, MX, TXT, and SRV records.

Zenmap Intense Scan



Zenmap Intense Scan against 10.0.0.0/24. The visible output shows the scan processing the address range and reporting many addresses as down.

Nikto web reconnaissance



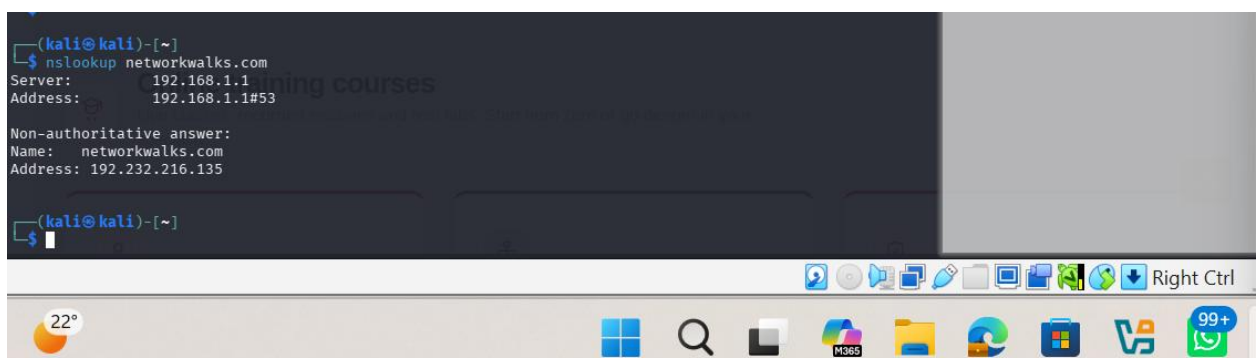
```
(kali@kali)-[~]
$ nikto -h networkwalks.com
- Nikto v2.6.0

+ Target IP: 192.232.216.135
+ Target Hostname: networkwalks.com
+ Target Port: 80
+ Platform: Unknown
+ Start Time: 2026-09-18 05:58:09 (GMT-4)

+ Server: Apache
+ ERROR: Failed to check for updates: 403
+ [999100] /: Uncommon header(s) 'x-endurance-cache-level' found, with contents: .
+ [999100] /: Uncommon header(s) 'x-nginx-cache' found, with contents: WordPress.
+ [999100] /: Uncommon header(s) 'x-redirect-by' found, with contents: WordPress - Really Simple Security
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options
clear
+ /dump.pem sent cookie: __wpdm_client=e53acd5a9bbf49e1745dbefe8838574; path=/; domain=networkwalks.com; HttpOnly
+ /site.pem sent cookie: __wpdm_client=ed901e84471b39dcef2ace1846552480; path=/; domain=networkwalks.com; HttpOnly
+ /database.pem sent cookie: __wpdm_client=4f1a3c0edfcd9f9f6fbfa285984b46d4; path=/; domain=networkwalks.com; HttpOnly
+ /com.tar.gz sent cookie: __wpdm_client=6ac9022ba24f64ff60436a36416837cc; path=/; domain=networkwalks.com; HttpOnly
+ /dump.tar.gz sent cookie: __wpdm_client=a0e64e6c0a5534b56129236046d5a7b1; path=/; domain=networkwalks.com; HttpOnly
+ /192.232.216.135.bz2 sent cookie: __wpdm_client=6aca719c409bd0206f1f76b1a098e9; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalks.com.gz sent cookie: __wpdm_client=c0e10dbad03fa06882a915b18d241064; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalkscom.tar.bz2 sent cookie: __wpdm_client=7f81acfd2de4ac4faddf00224b0c9a5c; path=/; domain=networkwalks.com; HttpOnly
+ /com.egg sent cookie: __wpdm_client=1b100dd82b8a9cb4ed840e367c05b5bd; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalkscom.tar.bz2 sent cookie: __wpdm_client=d85ace6b60c04607efd30bb5923e9a1a; path=/; domain=networkwalks.com; HttpOnly
+ /database.zip sent cookie: __wpdm_client=e694141452c36e4437039e4428a4754c; path=/; domain=networkwalks.com; HttpOnly
+ /192.232.216.135.jks sent cookie: __wpdm_client=6834a4eabbbcb108a0bb8eddb03036c61f; path=/; domain=networkwalks.com; HttpOnly
+ /dump.egg sent cookie: __wpdm_client=4249c2e15a1629583231a93357af88af; path=/; domain=networkwalks.com; HttpOnly
+ /archive.cer sent cookie: __wpdm_client=d6708dae92b25db8c4a9e2de4251fbca; path=/; domain=networkwalks.com; HttpOnly
+ /network.lzma sent cookie: __wpdm_client=02f63162127945d583924c84102279f4; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalks.com.bz2 sent cookie: __wpdm_client=c3778613a0c6bec85b6791f8885a5d53; path=/; domain=networkwalks.com; HttpOnly
+ /archive.tar sent cookie: __wpdm_client=abb16add883f7d953dd687d3955698c3; path=/; domain=networkwalks.com; HttpOnly
+ /backup.zip sent cookie: __wpdm_client=184bae4307a70a824089c4b04f7565b7; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalkscom.tar.lzma sent cookie: __wpdm_client=b8b98a4bd42c5b612e9933891f9feb59; path=/; domain=networkwalks.com; HttpOnly
+ /archive.jks sent cookie: __wpdm_client=c043561883984e7ee89864d0fc63e898; path=/; domain=networkwalks.com; HttpOnly
+ /com.tgz sent cookie: __wpdm_client=b6ffd004fe636558447148ee3aa66e97; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalks.com.tgz sent cookie: __wpdm_client=dce0ac6b550f570b09acbd404dc269f9; path=/; domain=networkwalks.com; HttpOnly
+ /site.tar sent cookie: __wpdm_client=3e7febaf39b0232875a2c9788dd28f32; path=/; domain=networkwalks.com; HttpOnly
+ /com.tar sent cookie: __wpdm_client=89be9c0843edba153e10b04388d36ca0; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalks.com.tar.bz2 sent cookie: __wpdm_client=d6cb36d04876f79d510a6b46338787ee; path=/; domain=networkwalks.com; HttpOnly
+ /backup.egg sent cookie: __wpdm_client=403dfddcef8f8e00a47acbb8e074af583; path=/; domain=networkwalks.com; HttpOnly
+ /dump.war sent cookie: __wpdm_client=7d472cc8dcac9c0c09bfb16de73121e; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalkscom.tar.lzma sent cookie: __wpdm_client=3f1ade906035be9e200b3503e3e6806b; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalkscom.tgz sent cookie: __wpdm_client=840cb721fdd255e07afcb3ac1ba24bec; path=/; domain=networkwalks.com; HttpOnly
+ /com.lzma sent cookie: __wpdm_client=5bfee6d393ef36596a0051638b58b0b7; path=/; domain=networkwalks.com; HttpOnly
+ /site.tgz sent cookie: __wpdm_client=c9e4dala76e656a9265e5ae79a51de56; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalkscom.tar sent cookie: __wpdm_client=82d10d6c0df7d9c62a66ebb4bbef2d9f; path=/; domain=networkwalks.com; HttpOnly
+ /networkwalks.com.zip sent cookie: __wpdm_client=5d20d3628c313434d17be9a5630cea29; path=/; domain=networkwalks.com; HttpOnly
```

Nikto 2.6.0 output for 192.232.216.135:80. The output reports Apache, WordPress-related headers, missing recommended security headers, and responses from multiple archive-style paths.

nslookup verification



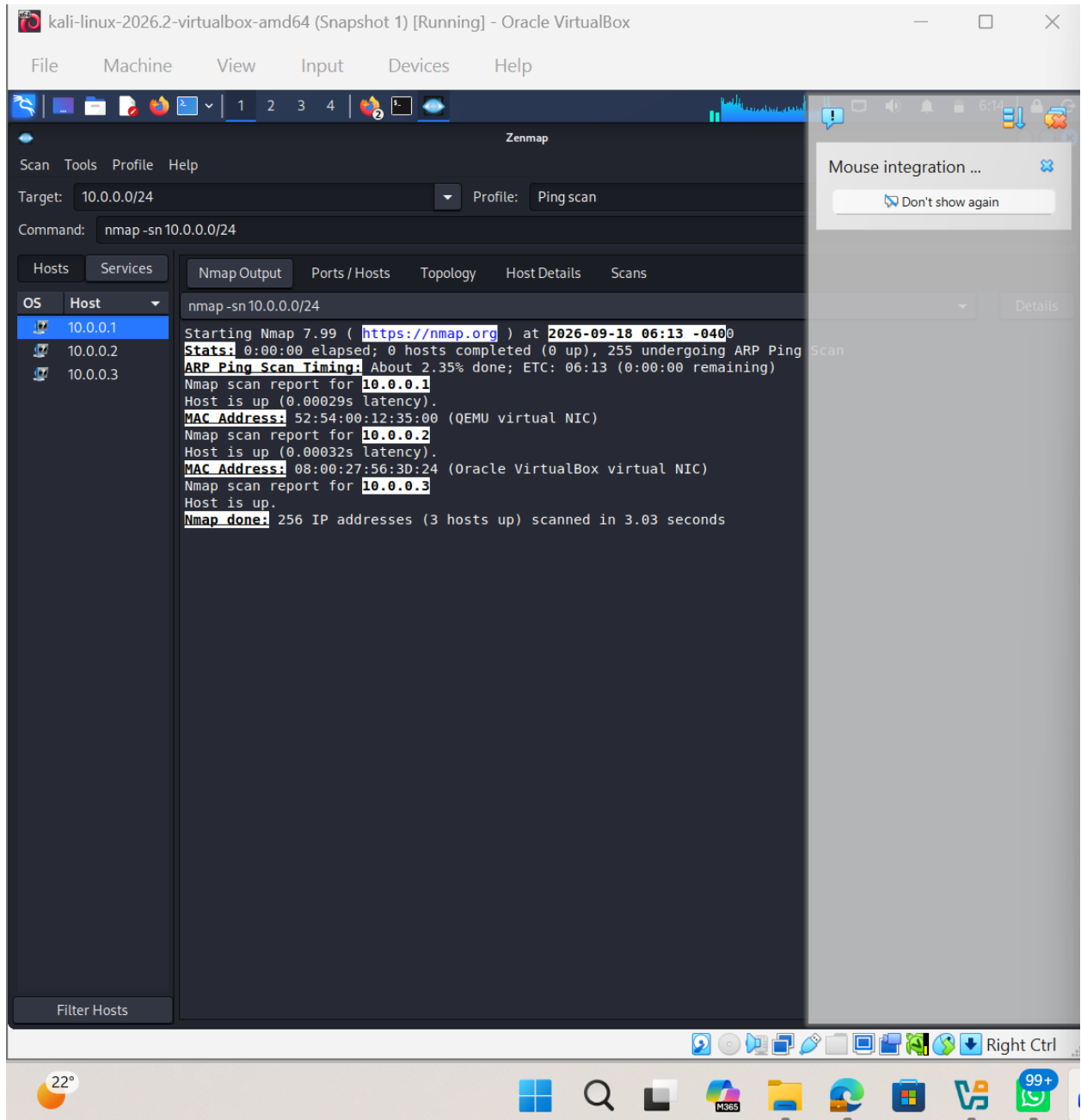
```
(kali@kali)-[~]
$ nslookup networkwalks.com
Server: 192.168.1.1
Address: 192.168.1.1#53

Non-authoritative answer:
Name: networkwalks.com
Address: 192.232.216.135

(kali@kali)-[~]
$
```

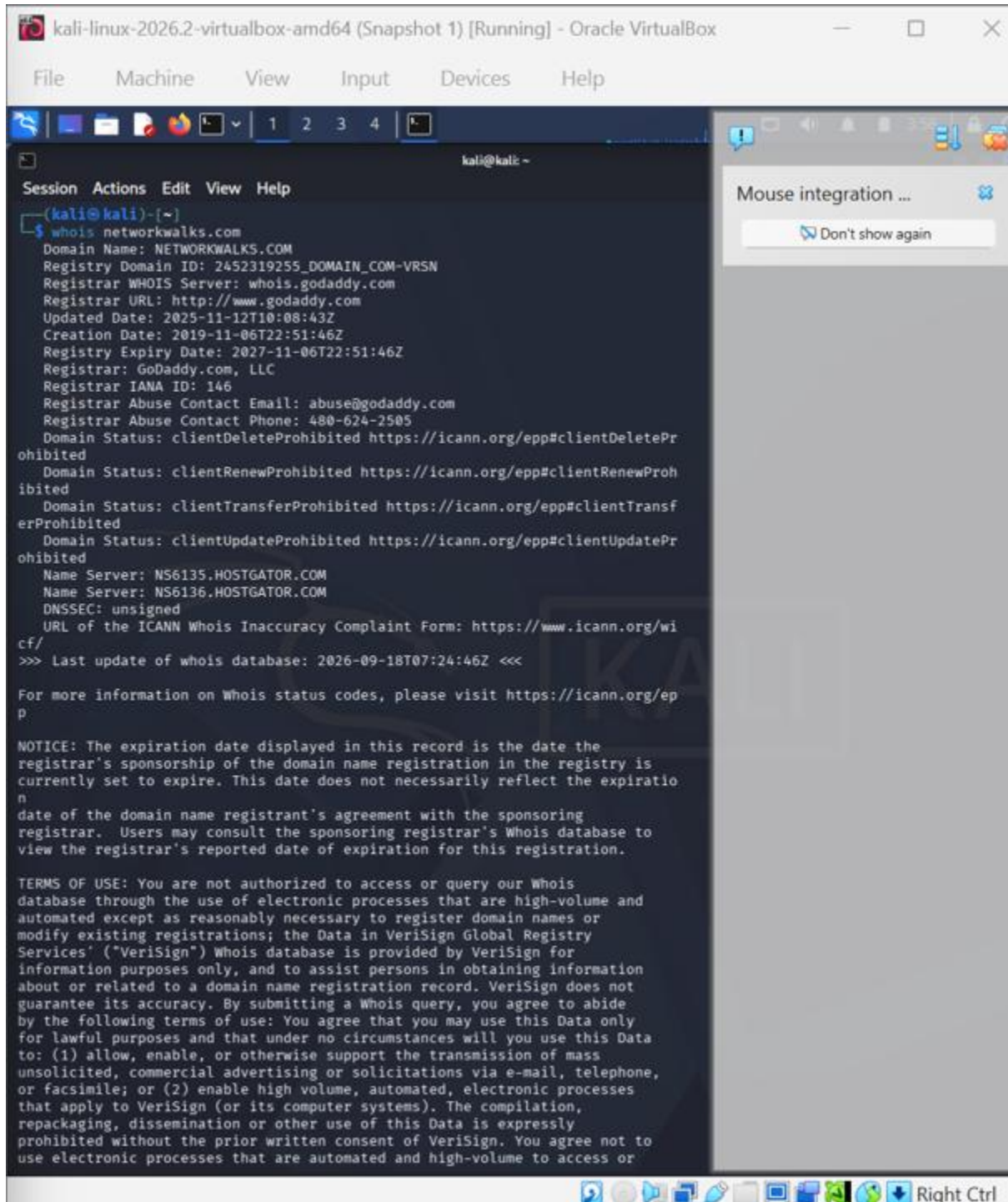
nslookup used 192.168.1.1 as the resolver and returned networkwalks.com at 192.232.216.135.

Zenmap Ping Scan



Zenmap Ping Scan against 10.0.0.0/24. The output reports 10.0.0.1, 10.0.0.2, and 10.0.0.3 as up.

WHOIS networkwalks.com output



```
kali@kali: ~  
$ whois networkwalks.com  
Domain Name: NETWORKWALKS.COM  
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.godaddy.com  
Registrar URL: http://www.godaddy.com  
Updated Date: 2025-11-12T10:08:43Z  
Creation Date: 2019-11-06T22:51:46Z  
Registry Expiry Date: 2027-11-06T22:51:46Z  
Registrar: GoDaddy.com, LLC  
Registrar IANA ID: 146  
Registrar Abuse Contact Email: abuse@godaddy.com  
Registrar Abuse Contact Phone: 480-624-2505  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeletePr  
ohibited  
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProh  
ibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransf  
erProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdatePr  
ohibited  
Name Server: NS6135.HOSTGATOR.COM  
Name Server: NS6136.HOSTGATOR.COM  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wi  
cf/  
>>> Last update of whois database: 2026-09-18T07:24:46Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/ep  
p  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiratio  
n  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for  
information purposes only, and to assist persons in obtaining information  
about or related to a domain name registration record. VeriSign does not  
guarantee its accuracy. By submitting a Whois query, you agree to abide  
by the following terms of use: You agree that you may use this Data only  
for lawful purposes and that under no circumstances will you use this Data  
to: (1) allow, enable, or otherwise support the transmission of mass  
unsolicited, commercial advertising or solicitations via e-mail, telephone,  
or facsimile; or (2) enable high volume, automated, electronic processes  
that apply to VeriSign (or its computer systems). The compilation,  
repackaging, dissemination or other use of this Data is expressly  
prohibited without the prior written consent of VeriSign. You agree not to  
use electronic processes that are automated and high-volume to access or
```

Zenmap Quick Scan Plus Output

The screenshot shows the Zenmap interface running in a Kali Linux virtual machine. The target is 10.0.0/24, and the profile is Quick scan plus. The command is nmap -sV -T4 -O -F --version-light 10.0.0/24. The output is displayed in the Nmap Output tab, showing results for 10.0.0.1 and 10.0.0.2.

Hosts: 10.0.0.1, 10.0.0.2, 10.0.0.3

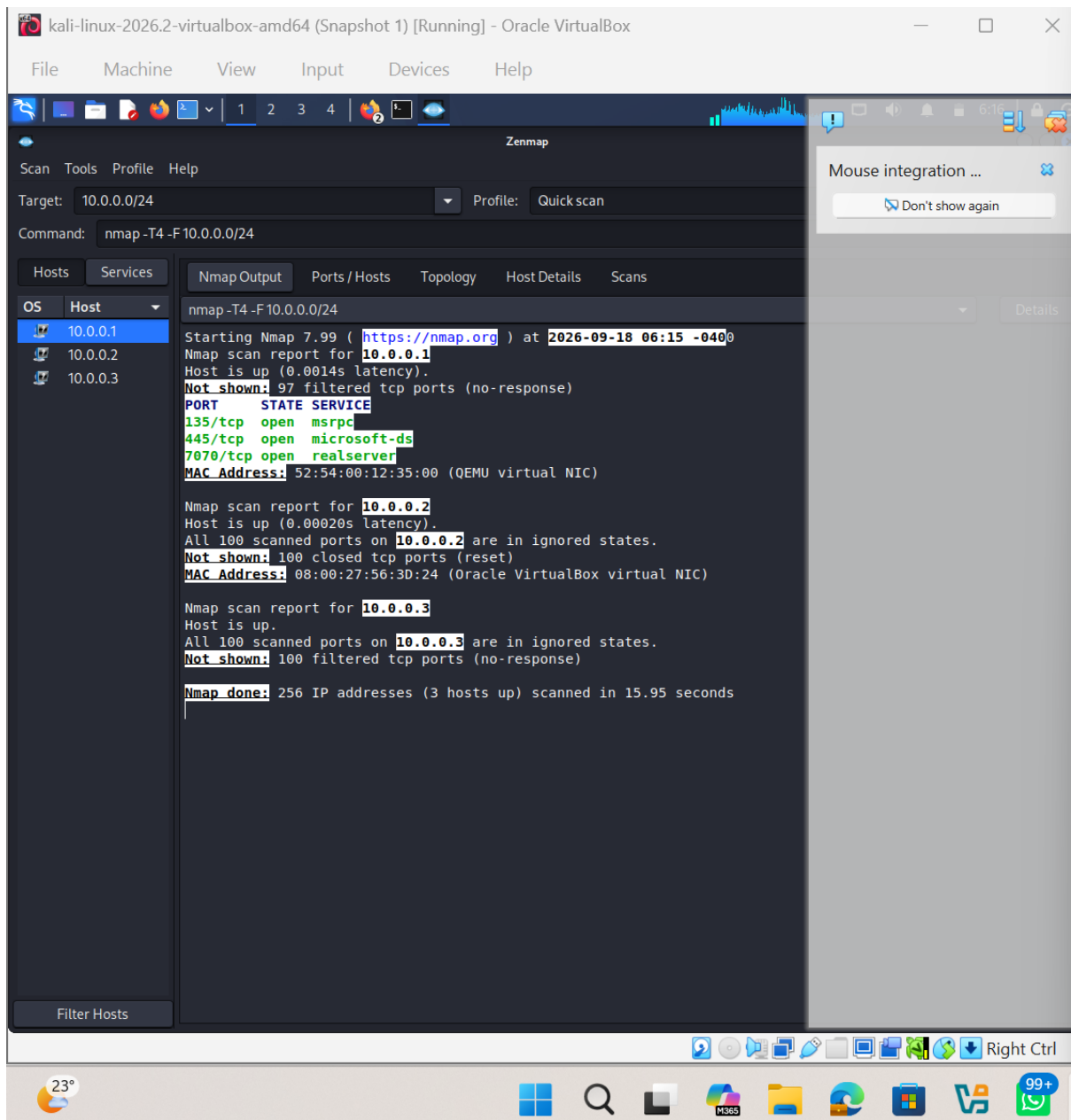
Nmap Output:

```
nmap -sV -T4 -O -F --version-light 10.0.0/24

Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 06:17 -0400
Nmap scan report for 10.0.0.1
Host is up (0.0014s latency).
Not shown: 97 filtered tcp ports (no-response)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
445/tcp    open  microsoft-ds?
7070/tcp   open  ssl/realserver?
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: VoIP adapter|bridge|general purpose|printer|switch
Running (JUST GUESSING): AT&T embedded (95%), Oracle Virtualbox (90%), Slirp (90%), QEMU (89%), HP embedded (86%), Minolta embedded (85%), NEC embedded (85%), Cisco embedded (85%)
OS CPE: cpe:/a:oracle:vm virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu cpe:/h:cisco:sg 500
Aggressive OS guesses: AT&T BGW210 voice gateway (95%), Oracle Virtualbox Slirp NAT bridge (90%), QEMU user mode network gateway (89%), HP 9100c Digital Sender printer (J3113A) (86%), Minolta Di550 laser printer (85%), NEC SuperScript printer (85%), Cisco SG 500 switch (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for 10.0.0.2
Host is up (0.00042s latency).
All 100 scanned ports on 10.0.0.2 are in ignored states.
Not shown: 100 closed tcp ports (reset)
MAC Address: 08:00:27:56:3D:24 (Oracle VirtualBox virtual NIC)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: 2N Helios IP VoIP doorbell (96%), Advanced Illumination DCS-100E lighting controller (96%), AudioControl D3400 network amplifier (96%), British Gas GS-Z3 data logger (96%), Chamberlain myQ garage door opener (96%), Daikin DKN Cloud Wi-Fi Adaptor (96%), Daysequerra M4.2SI radio (96%), Denver Electronics AC-5000W MK2 camera (96%), Eve Cam (lwIP 2.1.0 - 2.2.0) (96%), Fatek FBs-CBEH PLC Ethernet communication board (96%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop
```

Zenmap Quick scan Output



Curl -I networkwalks.com output

```
(kali@kali) [~]
$ wafw00f networkwalks.com

  Woof!

  ~ WAFW00F : v2.4.2 ~
  The Web Application Firewall Fingerprinting Toolkit

  [*] Checking https://networkwalks.com
  [+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
  [-] Number of requests: 2

(kali@kali) [~]
$ curl -I networkwalks.com
HTTP/1.1 301 Moved Permanently
Date: Fri, 18 Sep 2026 09:11:59 GMT
Server: Apache
Permissions-Policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
Expires: Fri, 18 Sep 2026 10:11:59 GMT
Cache-Control: max-age=3600
X-Redirect-By: WordPress - Really Simple Security
Set-Cookie: __wpdm_client=a62ff8cecc3e0e9001537d915cda7e5b; path=/; domain=networkwalks.com; HttpOnly
Upgrade: h2,h2c
Connection: Upgrade
Location: https://networkwalks.com/
Referrer-Policy: no-referrer-when-downgrade
X-Endurance-Cache-Level: 0
X-nginx-cache: WordPress
Content-Type: text/html; charset=UTF-8
```

Whatweb networkwalks.com

```
(kali@kali) [~]
$ whatweb networkwalks.com
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__wpdm_client], Country[UNITED STATES][US], HTTPServer[Apache], HttpOnly[permissions-policy,x-redirect-by,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache]
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.1], Cookies[__wpdm_client], Country[UNITED STATES][US], Email[info@networkwalks.com], JQuery[3.7.1], MetaGenerator[WordPress 7.1.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[4684NR-IPIB&pidnVar2=50511&pvtVar2=76&scvVar2=12,app], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.1]

(kali@kali) [~]
$
```